Factorial and Fibonacci Assignment

Factorial

Pseudocode:

```
START
INPUT n
IF n < 0 THEN
    PRINT "Invalid number"
ELSE
    factorial ← 1
    FOR i ← 1 TO n DO
        factorial ← factorial × i
    END FOR
    PRINT factorial
END IF
END
```

Python Code:

```
def factorial(n):
    if n < 0:
        return "Invalid number"

    result = 1
    for i in range(1, n + 1):
        result *= i

    return result
```

Fibonacci

Pseudocode:

```
START
INPUT n
SET a ← 0, b ← 1
PRINT a, b

FOR i ← 2 TO n-1 DO
    next ← a + b
    PRINT next
    a ← b
    b ← next
END FOR
END
```

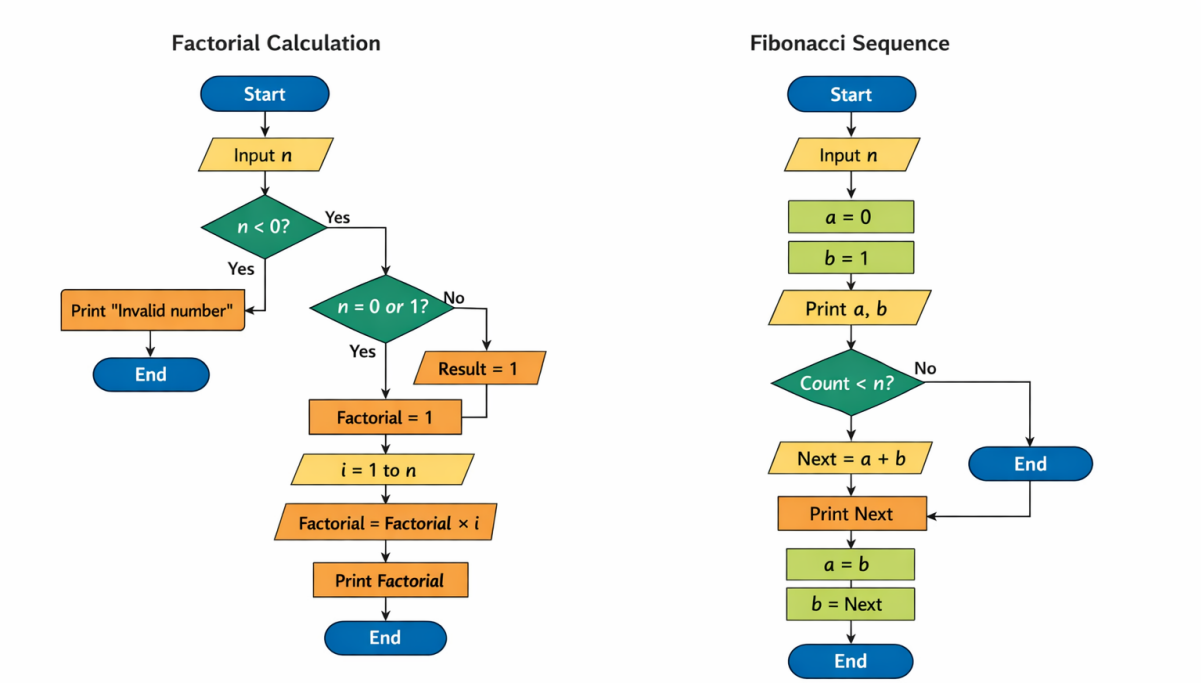
Python Code:

```
def fibonacci(n):
    if n <= 0:
        return "Invalid input"

    sequence = []
    a, b = 0, 1

    for i in range(n):
        sequence.append(a)
        a, b = b, a + b
    return sequence
```

Flowcharts


Factorial Calculation
Start
Input n
n < 0?
Yes
Yes
Print "Invalid number"
End
n = 0 or 1?
No
Yes
Result = 1
Factorial = 1
i = 1 to n
Factorial = Factorial × i
Print Factorial
End
Fibonacci Sequence
Start
Input n
a = 0
b = 1
Print a, b
Count < n?
No
Next = a + b
End
Print Next
a = b
b = Next
End